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Believe it or not? effects of untrue online comments on the AI era

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Abstract. Online comments from previous customers play a crucial role in influencing potential customers' decisions in retail, food, and beverage markets. Nonetheless, some comments are unreliable, as they are not customers' true experiences but are paid for by sellers. The fake comment experiment conducted by Oobah Butler on TripAdvisor in 2017 drew attention from sellers and customers. Nonetheless, the situation may become even worse if AI can automatically perform such services, making it difficult to uncover the truth with mere human resources. Customers may overlook valuable comments or even leave the market if they are misled. In some online retailing platforms in China, it is not uncommon to find service providers offering “likes,” false comments, and even responding to services against negative comments from genuine customers. A mixed-methods approach to investigate untrue online comments and their associated reactions, blending digital analysis with qualitative insights. The method uses automated web scraping tools to collect a large dataset of product reviews, reviewer histories, and full response threads from major Chinese e-commerce platforms, with a focus on the hospitality market. This article consists of three parts. First, it examines the impacts of misinformation on online retailing and the food and beverage industries, from user dissatisfaction to potential market losses. Second, it estimates the cost and difficulty of removing misinformation. Finally, the authors analyze whether the government and regulatory bodies should legislate against these behaviors and suggest how legislators and platforms can enhance the reliability of online comments.

Keywords: Fake comments; imbalance of information; misrepresentation; online platform; online shopping;

Introduction

Online reviews, false reviews, and consumer decisions

Online reviews were originally an important source of information for consumers when making purchase decisions. Especially in service-consumption scenarios such as hotel reservations, consumers cannot truly experience the hotel rooms, services, hygiene, or location before checking in. Therefore, they often rely on the reviews left by other consumers to reduce uncertainty. In other words, the role of reviews is not only to “providing information” but also to influence consumers' overall judgment of whether the hotel is trustworthy, risky, and worth booking.

However, once reviews are manipulated by humans, they no longer merely represent consumers' true experiences; instead, they become a fabricated information environment. Wu *et al.* (2020) pointed out that

false online reviews are not isolated minor issues but rather problems that systematically affect consumers' judgments, platform trust, and market efficiency. Luca & Zervas (2016) also found that in businesses under greater competitive pressure, with weaker reputations, or those that have recently received negative evaluations, the motivation for review fraud is stronger. He *et al.* (2021) further explained that purchasing false reviews may temporarily increase product ratings, the number of reviews, and sales performance.

In the context of the rapid development of artificial intelligence, this issue has become more complex. In the past, fake reviews might have needed to be written manually, but now generative AI can quickly generate a large number of seemingly natural, positive, and complete reviews. The MAiDE-up hotel review dataset constructed by Ignat *et al.* (2025) includes both real human hotel reviews and fake hotel reviews generated by GPT-4. Compared with manually written false reviews, the uniqueness of AI-generated reviews lies in their ability to produce large amounts of text quickly and cost-effectively, making the output appear smooth and natural. As a result, consumers are more likely to encounter a review environment in which it is more difficult to distinguish true from false. For hotel booking platforms, the problem is not just "whether there are fake reviews" but, more importantly, whether, after consumers see these reviews, they change their trust in the hotel, risk assessment, and booking intention.

Assessing the authenticity of reviews: the first step in influencing consumers

Existing research has already noted that consumers do not simply passively accept review information. Instead, consumers will judge whether a review is genuine based on its language, details, tone, repetition, and credibility cues. Walther *et al.* (2023) noted that consumers use multiple cues to assess the authenticity of online reviews. For instance, whether the review contains specific details, whether it resembles a real experience, whether it is overly exaggerated, or whether it resembles an advertisement can all affect consumers' judgment of the review's authenticity. However, the focus of this study is not on whether consumers can accurately identify each false review, but rather on whether they form different "perceptions of authenticity" after encountering different types of reviews. Therefore, this paper takes "perception of review authenticity" as the core variable, because consumers' subjective feelings often have a greater impact on their subsequent trust, risk assessment, and booking intentions than the objective truthfulness of the reviews. Here, "perception of authenticity" refers to consumers' subjective judgment of whether the review is true, credible, or written by a real user.

False reviews usually have some suspicious characteristics, such as overly general expressions, overly positive tone, lack of specific accommodation details, or seeming more like promotional texts. These characteristics may reduce consumers' judgment of review authenticity. On the contrary, genuine reviews usually contain more natural descriptions of experiences and more specific details, making them easier for consumers to find credible. Therefore, this study proposes:

H1: Compared with genuine reviews, fake reviews will reduce consumers' perception of the authenticity of the reviews.

Assessing the authenticity and trustworthiness of reviews

Trust is a very crucial variable in online consumption. When consumers book a hotel, they are actually purchasing a future service experience. Since consumers cannot fully confirm in advance whether the hotel is really clean, quiet, convenient, and has good service, they need to form trust in the hotel through reviews. Furner *et al.* (2022) found in the hotel and Airbnb contexts that online reviews affect consumers' trust, and trust further influences purchase and recommendation intentions. The judgment of review authenticity directly affects the formation of trust. In other words, the judgment of review authenticity serves as the crucial psychological bridge that helps consumers move from "seeing the reviews" to "believing in the hotel". If consumers believe the reviews were left by real guests, they are more like reliable third-party information. Consumers are also more likely to believe that the advantages displayed on the hotel's page are

real. On the contrary, if consumers believe the reviews are fabricated or written by the business itself, they not only fail to increase trust but may also make consumers doubt whether the hotel has any problems. Pedersen and Li (2024) also pointed out that when consumers encounter identified false reviews, their perceived trust declines. That is to say, false reviews do not necessarily directly lead consumers to abandon a purchase, but they will first undermine consumers' trust in the source of information and the merchant. Based on this logic, this study believes that the higher the authenticity of the reviews, the stronger the consumers' trust in the hotel. Therefore, it is proposed:

H2: The higher the consumers' perception of the authenticity of the reviews, the stronger their trust in the hotel.

Assessing the authenticity of reviews and perceiving risks

Apart from trust, perceiving risks is also an important variable in consumer decision-making. Perceived risk refers to the degree to which consumers believe the chosen option may result in losses, trouble, or uncertain consequences. In hotel reservations, risks may include the room not meeting expectations, poor hygiene, an inconvenient location, poor service, difficulty obtaining a refund, or spending money only to have a dissatisfying experience. Wu *et al.* (2019) pointed out that false reviews can increase consumers' functional, time, and economic risks and further reduce their purchase intentions. That is to say, when consumers doubt the authenticity of the reviews, they will be more concerned about being misled. For example, if a hotel review is full of positive reviews but lacks details, consumers might think: "Are these positive reviews genuine? What if I make a reservation and then find it's not good?" This suspicion will increase perceived risks. Pedersen & Li (2024) also found that false reviews can affect consumers' perception of risk. The less credible the reviews are, the more difficult it is for consumers to judge the hotel's true quality, and the more likely they are to feel the booking outcome is uncontrollable. Therefore, there should be a negative relationship between perceived review authenticity and perceived risk. In the research framework of this article, perceived risk does not occur in isolation; rather, it is a psychological concern that arises among consumers after they doubt the authenticity of the reviews. Based on this, this study proposes:

H3: The higher the consumers' perception of the authenticity of the reviews, the lower the perceived risk.

Trust, risk perception, and hotel reservation intentions

The willingness to book is the direct manifestation of whether consumers are willing to choose a hotel after reading the reviews. Online reviews influence consumers' decisions and are ultimately reflected in their purchase or booking intentions. Which? The 2020 behavioral experiment showed that when low-quality products were accompanied by false-positive reviews, the probability of consumers choosing them increased significantly. This indicates that false reviews not only affect "ideas" but also may change the actual direction of choice. In hotel reservations, the higher the level of trust, the more likely consumers are to believe the hotel can provide a satisfactory experience and to have the intention to make a reservation. Furner *et al.* (2022) found that trust is an important predictor of purchase intention. In other words, consumers do not simply make reservations based on the number of reviews or high ratings; instead, they make reservations because the reviews lead them to believe the hotel is trustworthy. On the contrary, the higher the perceived risk, the more likely consumers are to hesitate, continue to compare other hotels, and even abandon the booking. Wu *et al.* (2019) pointed out that when false reviews increase consumers' risk perception, purchase intention declines. Pedersen and Li (2024) also demonstrated that perceived risk plays a significant role in the process where false reviews affect purchase intention. Therefore, this study proposes:

H4: The stronger consumers' trust in the hotel is, the greater their willingness to make a hotel reservation after reading the reviews will be.

H5: The higher the consumers' perceived risk is, the weaker their willingness to make a hotel reservation after reading the reviews will be.

The indirect impact mechanism of false reviews on booking intentions

Previous studies have shown that false reviews can influence consumers' choices, but this influence is not necessarily immediate. After consumers see false reviews, they may not immediately react with "I won't book" or "I will book" respectively. A more likely process is as follows: Consumers first judge whether the review is true; if they think it is not, they will reduce their trust in the hotel or increase their concerns about booking risk; finally, these psychological changes will affect their booking intention. This logic is also in line with the summary of Wu *et al.* (2020) on research on false reviews: The issue of false reviews not only concerns the truthfulness of the reviews themselves but also consumers' judgments, platform trust, and market outcomes. Walther *et al.* (2023) focused on how consumers judge the truthfulness of reviews, while Wu *et al.* (2019), Pedersen & Li (2024), and Furner *et al.* (2022) further supported the relationship chain of "truthfulness judgment - trust/risk - purchase intention". Therefore, this study aims not only to examine whether false reviews directly affect the willingness to make a reservation but also to further investigate their psychological mechanisms. Specifically, false reviews may first reduce consumers' perception of review authenticity and then, by lowering trust or increasing perceived risks, indirectly reduce willingness to make a hotel reservation. Based on this, this study proposes:

H6: False reviews will indirectly reduce the willingness to make a hotel reservation after reading the reviews by lowering the perceived authenticity of the reviews and further weakening trust.

H7: False reviews will indirectly reduce the willingness to make a hotel reservation after reading them by lowering the perceived authenticity of the reviews and further increasing perceived risks.

Research gaps and theoretical positioning of this study

Overall, existing studies have demonstrated that false reviews can influence consumers' judgment and purchasing behavior; there have also been discussions on how consumers can identify false reviews and how they affect trust, risk, and purchase intention (Pedersen & Li, 2024; Walther *et al.*, 2023; Wu *et al.*, 2019, 2020). However, existing research still leaves a question worthy of further investigation: In contexts where AI can generate false hotel reviews, in real hotel booking scenarios, how exactly do consumers progress from "seeing the reviews" to "changing their booking intentions"? The research gap of this article is not merely "whether false reviews will affect consumers", but "how AI-generated false hotel reviews can influence consumers' booking intentions through several psychological steps, such as authenticity judgment, trust, and risk". Rather than merely focusing on identifying false reviews, this study pays closer attention to the consumer decision-making process. Specifically, this study situates consumers in the context of hotel reservations, compares the effects of real reviews and AI-generated false reviews, and simultaneously examines the relationships among perceived review authenticity, trust, perceived risk, and booking intentions. Such a research design can more clearly illustrate that false reviews do not necessarily directly change consumers' final booking intentions but may first alter consumers' judgment of the review's authenticity, and then influence the booking decision through the psychological channels of trust and risk.

Discussion

Online reviews were originally an important tool to help consumers reduce information asymmetry. However, once the reviews are manipulated, what consumers see is no longer the true information about the product, but an artificially influenced information environment. Wu *et al.* (2020) summarized fake online reviews as a problem that systematically affects consumers' judgments, platform trust, and market efficiency, and it is not just scattered noise. That is to say, the problem with fake reviews is not simply that "some

statements are not true" but that they directly interfere with the purchase decisions consumers should make based on true information.

Regarding consumer decision-making, these papers can be connected into a very clear logical line. Wu *et al.* (2019) found that fake reviews increase consumers' functional, time, and economic risks, and that these risks can be so high as to lower purchase intention. Pedersen & Li (2024) further explained that when consumers see identified fake reviews, their perceived trust would decrease, their perceived quality would decline, their perceived risks would increase, and ultimately their purchase intention would also be affected. Furner *et al.* (2022), although not studying fake reviews themselves, reported experimental results that are very important because they showed that trust is an important predictor of purchase intention. When these several papers are viewed together, it can be explained that fake reviews would further change consumers' purchase intentions and decisions by influencing trust and perceived risk.

This influence is not abstract. The behavioral experiments conducted revealed that when low-quality products are accompanied by false positive reviews, the probability of consumers choosing these products significantly increases. If additional manipulations such as star rating manipulation, false text, and platform recommendation tags are added, the misleading effect becomes even stronger. Even if a reminder like "These reviews may not be genuine" is added to the page, it can only partially mitigate the deception and cannot completely eliminate its impact. This is very important because it indicates that consumers are influenced not only when "no signs of problems can be detected"; even when there are some suspicious signals on the page, as long as the overall presentation remains positive, consumers may still make choices that differ from their original intentions.

From the supply side, the act of merchants fabricating fake reviews is not accidental but driven by a clear economic motive. Luca & Zervas (2016) found that approximately 16% of restaurant reviews on Yelp are automatically filtered by the system as suspicious, and that review fraud is more likely to occur in businesses with weak reputations, few reviews, recent negative reviews, and greater competitive pressure. He *et al.* (2021)'s study of the fake review market on Amazon also indicates that purchasing fake reviews can indeed increase ratings, the number of reviews, and sales in the short term, but this effect is usually only temporary; once the purchase of fake reviews stops, the average rating of the product will decline, and the proportion of one-star negative reviews will increase. This shows that fake reviews do indeed change the product's image in consumers' eyes in the short term, and that consumers' initial choices were likely misled. The citation of these two articles is not intended to write an article titled "Why do merchants fake reviews?", but rather to illustrate that: The reason why fake reviews are worthy of study is that they do enter the market and do affect the information that consumers come into contact with.

However, this study does not intend to focus on "how to identify fake reviews". Walther *et al.* (2023) reviewed how consumers determine the authenticity of reviews, indicating that consumers use certain cues to assess whether a review is trustworthy; however, "whether consumers can detect suspicious reviews" and "whether these reviews will influence consumers' purchasing decisions" are not the same question. The former focuses on identification clues, while the latter focuses on behavioral consequences. For your study, the more important aspect is the latter, that is, whether consumers will change their original judgment after reading the reviews. This perspective is closer to consumer decision-making and is more suitable for directly incorporating into a questionnaire study.

Wu *et al.* (2020) specifically pointed out that many key questions regarding fake reviews are difficult to answer solely using existing platform data. The reason is simple: researchers usually find it challenging to prove with 100% certainty which reviews are definitely fake, and it is also difficult to directly observe the counterfactual result of "what consumers would have chosen if these fake reviews did not exist". Because the real platform data does not show this "what would have happened" situation, experiments and

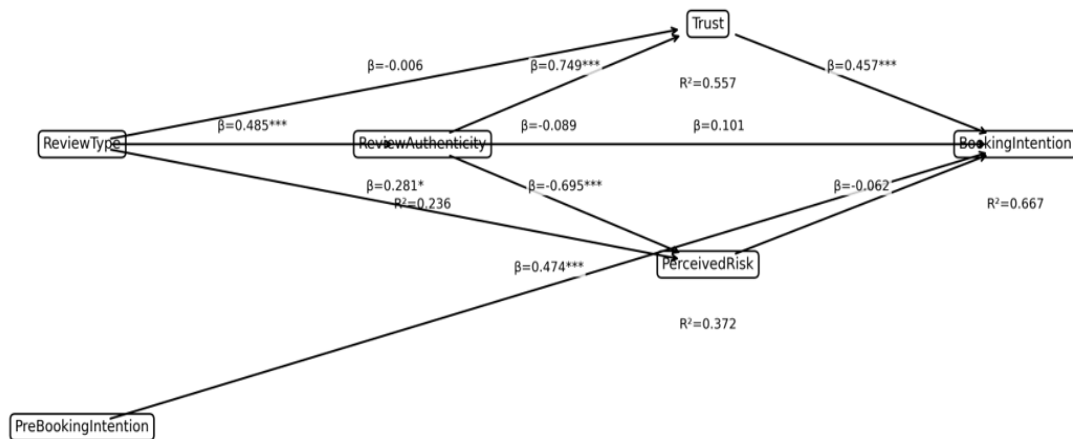
questionnaire experiments are particularly suitable. Wu *et al.* (2019) and Which? (2020) both demonstrated that as long as the respondents are placed in a specific shopping scenario and their risk perception, trust changes, and purchase intentions are measured, it is possible to effectively capture the mechanism by which fake reviews affect consumers. Therefore, the use of questionnaire or situation experiment methods in this study is reasonable: it can directly compare whether the respondents' purchase intentions have changed before and after reading the reviews, determine whether there is a change, whether they are more willing to buy, whether they are more hesitant, and whether they need more time to consider.

Based on these previous studies, this research focuses on "whether the purchase decision changes after consumers read the reviews", which is well-grounded. Wu *et al.* (2019), Pedersen & Li (2024), and Furner *et al.* (2022) can be used to support the relationship between trust, risk, and purchase intention; Which? (2020) can be used to support the claim that fake reviews can directly alter the selection outcome; Wu *et al.* (2020) can be used to support the reason for using questionnaires and situational experiments; Lucas & Zervas (2016) and He *et al.* (2021) can be used to illustrate that fake reviews do exist in the real market and do indeed shape the information environment that consumers are exposed to. Therefore, this study places consumers in a product browsing scenario, compares whether their judgments change before and after viewing the reviews, and simultaneously measures trust, risk, and purchase intention. This is a relatively complete and robust research path.

Methodology

Research design and experimental context

This study employs a situational questionnaire experiment to examine how the types of online hotel reviews affect consumers' perceptions of review authenticity, trust, perceived risks, and post-review booking intentions. The research context is set as online hotel booking. In this context, consumers typically rely on user-generated reviews to assess the credibility of hotel information and identify potential risks in their booking decisions. Therefore, the core issue of this study is not whether consumers can accurately identify false reviews, but rather how their judgments on the authenticity of hotel reviews, trust in the hotel, perceived risks, and booking intentions will change after they come into contact with different authenticity cues in hotel reviews.



Note: The numbers on the arrows represent the standardized path coefficients. R square indicates the degree to which the model explains the dependent variable.

FIGURE - The PLS-SEM structural model of the influence of comment types on the willingness to make a reservation after reading the comments.

This study prepared two versions of the review stimulus materials. The materials for the real review group used more natural, closer-to-personal-experience expressions, with more specific experience details and stronger credibility cues. The materials for the fake review group had a more overtly promotional tone and were more general, with weaker authenticity cues. Therefore, the review type was treated as an exogenous variable manipulated in the experiment, with the real review group coded as 1 and the fake review group as 0. To avoid the influence of the direction of the comments' emotions on the results, this study controlled both sets of comment materials to be positive hotel reviews. Only the authenticity cues of the comments were changed. The questionnaire process was divided into three stages. First, participants entered the hotel booking context and reported their pre-review booking intentions before reading the review materials. Second, participants read one of the versions of the online review materials. Third, participants evaluated the perceived authenticity of the review, trust, perceived risk, post-review booking intention, and several supplementary behavioral outcomes. Measuring pre-review booking intention before exposure to review materials helps distinguish consumers' original interest in the hotel from the psychological evaluation formed after reading the reviews.

Sample and data collection

Data was collected through online questionnaires. The front page of the questionnaire informed participants that this study is for academic purposes only, that all responses are anonymized, and that the data will be analyzed only in summary form. Participants first completed the informed consent and screening questions. Respondents entering the formal analysis needed to meet two conditions: being over 18 years old and having experience in online shopping, online hotel booking, or other online appointment activities. This study received 72 questionnaire responses. According to the screening criteria, 4 responses were excluded because the respondents did not report relevant online shopping or booking experience. The final valid sample comprised 68 participants, with 32 in the real review group and 36 in the fake review group. Given the small sample size, this study should be regarded as exploratory; however, this article still presents the measurement and structural model results in accordance with the complete PLS-SEM reporting framework to transparently display the measurement properties and structural relationships of the model. This study chose PLS-SEM because it is exploratory in nature, the sample size is relatively small, and the model includes multiple psychological variables and mediating paths. The research objective is also more focused on explaining and predicting consumers' booking intentions following review reading.

Measurement and variable operationalization

All multi-item constructs, except for review type, were measured on a five-point Likert scale, with 1 indicating "strongly disagree" and 5 indicating "strongly agree". The model included five core constructs: pre-review booking intention, perceived review authenticity, trust, perceived risk, and post-review booking intention. The review type was entered as a single binary experimental variable into the model. To reduce the possibility of mechanical responses, several reverse-scored items were included in the questionnaire; these items were reverse-scored according to $X_{reversed} = 6 - X$ before the formal analysis.

TABLE 1 - Sample screening process

Screening step	Sample size
Initial responses	72
After adult-status screening	72
After online shopping or booking experience screening	68

Note. The final sample used in the PLS-SEM analysis contained 68 valid responses.

Table 2 - Constructs and operationalization

Construct	Variable name	Items	Reverse-coded items	Role in the model
Review type	ReviewType	Single binary variable	Real = 1; fake = 0	Experimentally manipulated exogenous variable
Pre-review booking intention	PreBookingIntention	PRE1-PRE4	PRE4	Control for consumers' baseline booking tendency
Perceived review authenticity	ReviewAuthenticity	AUTH1-AUTH4	AUTH2, AUTH3	Credibility, genuineness, and accuracy of the reviews
Trust	Trust	TRUST1-TRUST4	TRUST4	Trust in the hotel after reading the reviews
Perceived risk	PerceivedRisk	RISK1-RISK4	None	Concerns about experience, monetary loss, and wrong decisions
Post-review booking intention	BookingIntention	BI1-BI7	BI5, BI7	Final booking intention after review exposure

Pre-review booking intention was used to control the consumers' original booking tendency. Perceived review authenticity indicated the degree to which respondents believed the review information was true, credible, and accurate. Trust indicated the respondents' confidence in the hotel after reading the review. Perceived risk indicated the respondents' concerns about the accommodation experience, financial loss, and incorrect booking decisions. Post-review booking intention indicated respondents' final booking intentions after reading the review materials. Decision changes and other supplementary behavioral outcomes were only used for descriptive analysis and were not included in the core structural model.

Data processing and analysis program

First, merge the data from the two questionnaire versions and generate binary comment-type variables. Second, rename the items listed in the questionnaire platform and match them to the corresponding measurement indicators. Third, recode the reverse items before estimating the construct. Fourth, the analysis follows the two-step logic commonly used in PLS-SEM research: first, evaluate the measurement model, and then evaluate the structural model. Specifically, the data analysis proceeds in the following sequence: "data cleaning, reverse item recoding, descriptive statistics, manipulation test, measurement model test, structural model test, Bootstrap mediation effect test, and IPMA analysis". This study uses a Python-based PLS-SEM workflow to estimate the model. All multi-item constructs are set to reflective. The latent variable scores are estimated from their respective indicators, and the relationships among latent variables are reported as standardized path coefficients. Statistical significance is assessed using 5000 bootstrap resamples. The measurement model is evaluated using outer loadings, Cronbach's alpha, rho_A, composite reliability, average variance extracted (AVE), Fornell-Larcker criterion, and HTMT ratio. The structural model is evaluated using internal VIF, path coefficients, explanatory power (R-squared), predictive relevance (Q-squared predict), effect size (F-squared), bootstrap indirect effect, and importance-performance diagram analysis (IPMA). In IPMA, the post-comment reservation intention is set as the target construct; the importance is represented by the total effect, and the performance value is recalibrated to the 0-100 range.

Program control of common method bias and collinearity test

This study employs various procedural methods to reduce common method bias. Specifically, the questionnaire was designed for anonymous submission, presented the experimental materials and core measurement items in separate pages, included reverse questions and attention checks, and during the statistical analysis stage, VIF was used to detect potential multicollinearity issues in order to reduce the impact of common method bias on the research results.

Table 3 - Descriptive statistics of core constructs

Construct	N	Mean	SD	Min	Max
Pre-review booking intention	68	2.846	0.768	1.000	4.250
Perceived review authenticity	68	2.941	0.781	1.000	4.750
Trust	68	3.213	0.682	1.750	5.000
Perceived risk	68	3.754	0.710	2.000	5.000
Post-review booking intention	68	3.284	0.734	1.286	5.000

Note. All multi-item constructs were measured on five-point Likert scales.

TABLE 4 - Mean comparison between the real-review and fake-review groups

Construct	Real mean	Fake mean	Mean difference	t-value	p-value	Cohen's <i>d</i>
Pre-review booking intention	2.891	2.806	0.085	0.449	.655	0.110
Perceived review authenticity	3.336	2.590	0.746	4.520	< .001	1.080
Trust	3.477	2.979	0.497	3.197	.002	0.778
Perceived risk	3.719	3.785	-0.066	-0.385	.702	-0.092
Post-review booking intention	3.397	3.183	0.215	1.209	.231	0.294

Note. Positive mean differences indicate higher scores in the real-review condition.

This study also conducts a statistical test for collinearity. Most VIF values are below the stricter 3.3 standard. The VIF for perceived comment authenticity in predicting post-comment reservation intention is 3.706, higher than 3.3 but still below the more lenient 5.0 standard. Therefore, collinearity is not considered to pose a serious threat to structural estimation. However, when interpreting the structural model, note the relatively close relationship among comment authenticity, trust, and reservation intention.

Results

Descriptive statistics and manipulation test

Table 3 reports the descriptive statistics of the core constructs. The means of perceived comment authenticity, trust, perceived risk, and post-comment reservation intention are 2.941, 3.213, 3.754, and 3.284, respectively. Table 4 reports the mean comparison between the two experimental groups. The perceived comment authenticity of the real review group was significantly higher than that of the fake review group (mean difference = 0.746, $p < .001$, Cohen's $d = 1.080$). The trust level of the real review group was also significantly higher (mean difference = 0.497, $p = .002$, Cohen's $d = 0.778$). In contrast, there were no significant differences in pre-review reservation intention, perceived risk, and post-review reservation intention between the two groups. This indicates that the experimental manipulation mainly functions through authenticity-related judgments and trust-related judgments, rather than directly causing changes in the mean values of downstream behavioral intentions.

Evaluation of the measurement model

First, evaluate the internal consistency, reliability, and convergent validity of the reflective measurement model. As shown in Table 5, the combined reliability of multi-item constructs ranges from 0.840 to 0.900, all exceeding the recommended standard of 0.70. The AVE values range from 0.564 to 0.693, all exceeding the standard of 0.50. The Cronbach's alpha for pre-review reservation intention is slightly below 0.70, but the construct's reliability and AVE are acceptable. Therefore, for this exploratory study, the reliability and convergent validity evidence are acceptable.

TABLE 5 - Reliability and convergent validity

Construct	Items	Cronbach's alpha	rho_A	CR	AVE
Review type	1	-	-	1.000	1.000
Pre-review booking intention	4	0.695	0.833	0.840	0.610
Perceived review authenticity	4	0.853	0.902	0.900	0.693
Trust	4	0.737	0.843	0.844	0.584
Perceived risk	4	0.825	0.891	0.888	0.669
Post-review booking intention	7	0.865	0.898	0.898	0.564

Note. Review type is a single binary experimental variable; internal consistency statistics are therefore not interpreted for this variable.

Table 6 presents the outer loadings. Most indicators exceed the recommended standard of 0.708. A few indicators have lower loadings, including PRE4, TRUST4, RISK2, BI5, and BI7. Since these items reflect the theoretical related aspects of the constructs and deleting items may narrow the coverage of the measurement content in a small sample exploratory study, these items were retained in the main analysis. However, further studies should revise and re-examine these indicators in a larger sample.

Table 6 - Outer loadings

Construct	Indicator	Outer loading	Loading squared	Assessment
Review type	ReviewType	1.000	1.000	Above 0.708
Pre-review booking intention	PRE1	0.906	0.821	Above 0.708
Pre-review booking intention	PRE2	0.921	0.848	Above 0.708
Pre-review booking intention	PRE3	0.859	0.739	Above 0.708
Pre-review booking intention	PRE4	0.178	0.032	Below 0.708
Perceived review authenticity	AUTH1	0.891	0.794	Above 0.708
Perceived review authenticity	AUTH2	0.763	0.581	Above 0.708
Perceived review authenticity	AUTH3	0.814	0.662	Above 0.708
Perceived review authenticity	AUTH4	0.857	0.735	Above 0.708
Trust	TRUST1	0.824	0.679	Above 0.708
Trust	TRUST2	0.921	0.849	Above 0.708
Trust	TRUST3	0.728	0.530	Above 0.708
Trust	TRUST4	0.529	0.280	Below 0.708
Perceived risk	RISK1	0.840	0.706	Above 0.708
Perceived risk	RISK2	0.609	0.371	Below 0.708
Perceived risk	RISK3	0.925	0.856	Above 0.708
Perceived risk	RISK4	0.862	0.744	Above 0.708
Post-review booking intention	BI1	0.819	0.671	Above 0.708
Post-review booking intention	BI2	0.829	0.687	Above 0.708
Post-review booking intention	BI3	0.729	0.531	Above 0.708
Post-review booking intention	BI4	0.843	0.711	Above 0.708
Post-review booking intention	BI5	0.579	0.335	Below 0.708
Post-review booking intention	BI6	0.866	0.749	Above 0.708
Post-review booking intention	BI7	0.515	0.265	Below 0.708

Note. Indicators below 0.708 were retained because of conceptual coverage and the exploratory nature of the study.

TABLE 7 - Fornell-Larcker discriminant validity assessment

	RT	PreBI	PRA	TR	PR	PostBI
RT	1.000	0.043	0.485	0.358	-0.056	0.148
PreBI	0.043	0.781	0.081	0.337	-0.026	0.635
PRA	0.485	0.081	0.833	0.746	-0.558	0.473
TR	0.358	0.337	0.746	0.764	-0.377	0.685
PR	-0.056	-0.026	-0.558	-0.377	0.818	-0.299
PostBI	0.148	0.635	0.473	0.685	-0.299	0.751

Note. Diagonal values represent the square root of AVE; off-diagonal values represent construct correlations. RT = review type; PreBI = pre-review booking intention; PRA = perceived review authenticity; TR = trust; PR = perceived risk; PostBI = post-review booking intention.

Table 8. HTMT discriminant validity assessment

	RT	PreBI	PRA	TR	PR	PostBI
RT	1.000	-	-	-	-	-
PreBI	-	1.000	0.248	0.524	0.287	0.788
PRA	-	0.248	1.000	0.912	0.656	0.513
TR	-	0.524	0.912	1.000	0.494	0.820
PR	-	0.287	0.656	0.494	1.000	0.347
PostBI	-	0.788	0.513	0.820	0.347	1.000

Note. HTMT values involving the single-item review type variable are not interpreted. The HTMT value between perceived review authenticity and trust exceeds the conservative threshold.

TABLE 9 - Inner VIF values

Target construct	Predictor	VIF	Assessment
Perceived review authenticity	Review type	1.000	Below 3.3
Trust	Review type	1.308	Below 3.3
Trust	Perceived review authenticity	1.308	Below 3.3
Perceived risk	Review type	1.308	Below 3.3
Perceived risk	Perceived review authenticity	1.308	Below 3.3
Post-review booking intention	Review type	1.435	Below 3.3
Post-review booking intention	Pre-review booking intention	1.220	Below 3.3
Post-review booking intention	Perceived review authenticity	3.706	Between 3.3 and 5.0
Post-review booking intention	Trust	2.752	Below 3.3
Post-review booking intention	Perceived risk	1.602	Below 3.3

Note. All VIF values were below 5.0.

Discriminant validity is evaluated using the Fornell-Larcker criterion and HTMT. The Fornell-Larcker results in Table 7 show that the square root of the AVE for each multi-item construct is generally higher than its correlations with other constructs. The HTMT results are reported in Table 8. Most HTMT values are within the acceptable range. However, the HTMT value between Perceived Comment Authenticity and Trust is 0.912, which exceeds the more conservative threshold. This indicates that, in the

context of hotel reviews, respondents may view judgments of comment authenticity and trust as highly correlated psychological evaluations. Therefore, the structural paths involving these two constructs need to be interpreted with caution.

Structural model evaluation

Before evaluating the structural paths, this study used the internal VIFs to assess collinearity among the predictor variables. As shown in Table 9, all VIF values were below 5.0, indicating that collinearity would not cause serious problems for the model estimation. The VIF for the perception of comment authenticity in predicting post-comment reservation intention was 3.706, indicating that this variable was only moderately correlated with other predictors, but did not reach the level of severe multicollinearity.

Table 10 reports the estimated results of the structural paths. The perception of comment authenticity has a significant positive effect on the comment type predictor variable ($\beta = 0.485, p < .001$), indicating that the real comment group is perceived as more authentic than the false comment group. The perception of comment authenticity has a strong positive effect on trust ($\beta = 0.749, p < .001$), and a strong negative effect on perception of risk ($\beta = -0.695, p < .001$). These results indicate that authenticity cues are closely related to higher trust and lower perception of risk. When predicting post-comment reservation intention, pre-comment reservation intention ($\beta = 0.474, p < .001$) and trust ($\beta = 0.457, p < .001$) are significant positive predictors. In contrast, the direct effect of the perception of comment authenticity on post-comment reservation intention is not significant ($\beta = 0.101, p = .479$), and the direct effect of perception of risk is also not significant ($\beta = -0.062, p = .554$). Comment type cannot directly predict post-comment reservation intention. Therefore, the structural model indicates that comment authenticity primarily affects reservation intention through the trust mechanism, rather than via a direct behavioral path. The positive coefficient for Comment Type on Perceived Risk should be interpreted with caution. The mean comparison shows no significant difference in Perceived Risk between the genuine comment group and the fake comment group; meanwhile, there is a strong negative relationship between Perceived Comment Authenticity and Perceived Risk. Therefore, this significant positive coefficient is more appropriately understood as a suppressive effect that occurs after controlling for Perceived Comment Authenticity, rather than suggesting that genuine comments will substantially increase Perceived Risk.

TABLE 10. Structural path results

Structural path	beta	t-value	p-value	95% CI	Significant
Review type -> Review authenticity	0.485	6.195	< .001	[0.323, 0.633]	Yes
Review type -> Trust	-0.006	-0.063	.950	[-0.190, 0.176]	No
Review authenticity -> Trust	0.749	11.302	< .001	[0.622, 0.878]	Yes
Review type -> Perceived risk	0.281	2.542	.011	[0.064, 0.487]	Yes
Review authenticity -> Perceived risk	-0.695	-7.790	< .001	[-0.847, -0.500]	Yes
Review type -> Post-review booking intention	-0.089	-1.002	.317	[-0.258, 0.095]	No
Pre-review booking intention -> Post-review booking intention	0.474	5.960	< .001	[0.308, 0.620]	Yes
Review authenticity -> Post-review booking intention	0.101	0.708	.479	[-0.225, 0.346]	No
Trust -> Post-review booking intention	0.457	4.207	< .001	[0.267, 0.693]	Yes
Perceived risk -> Post-review booking intention	-0.062	-0.592	.554	[-0.271, 0.140]	No

Note. The results are based on 5,000 bootstrap resamples. Significant = Yes indicates $p < .05$.

TABLE 11 - Explanatory power and predictive relevance

Endogenous construct	R-squared	Adjusted R-squared	Q-squared predict	RMSE	MAE
Perceived review authenticity	0.236	0.224	0.216	0.886	0.700
Trust	0.557	0.543	0.546	0.674	0.524
Perceived risk	0.372	0.353	0.344	0.810	0.654
Post-review booking intention	0.667	0.640	0.598	0.634	0.505

Note. Positive *Q*-squared predict values indicate predictive relevance for the endogenous constructs.

TABLE 12 - *F*-squared effect sizes

Target construct	Predictor	F-squared	Effect size
Perceived review authenticity	Review type	0.308	Medium
Trust	Review type	0.000	Very small
Trust	Perceived review authenticity	0.969	Large
Perceived risk	Review type	0.096	Small
Perceived risk	Perceived review authenticity	0.588	Large
Post-review booking intention	Review type	0.016	Very small
Post-review booking intention	Pre-review booking intention	0.554	Large
Post-review booking intention	Perceived review authenticity	0.008	Very small
Post-review booking intention	Trust	0.228	Medium
Post-review booking intention	Perceived risk	0.007	Very small

Note. Effect size interpretation follows conventional benchmarks: 0.02 = small, 0.15 = medium, and 0.35 = large.

The model explains 23.6% of the variance in Perceived Comment Authenticity, 55.7% in Trust, 37.2% in Perceived Risk, and 66.7% in Post-Booking Intent. The R-squared value for Post-Booking Intent indicates that the model has strong explanatory power for the final outcome variable. Additionally, all Q-squared predicted values are positive, supporting the model's predictive relevance.

The F-squared results further indicate that perceived comment authenticity has a significant effect on both trust and perceived risk. Pre-comment booking intention has a significant effect on post-comment booking intention, whereas trust has a moderate effect. After including other predictor variables, the direct effects of comment type, perceived comment authenticity, and perceived risk on post-comment booking intention are all very small.

Mediation effect analysis

This study uses bootstrap indirect-effect tests to examine the mediating relationship. As shown in Table 13, the indirect effect of comment type through perceived comment authenticity on post-comment booking intention is not significant. However, the chain indirect effect from comment type to perceived comment authenticity, to trust, and finally to post-comment booking intention is significant (effect = 0.166, $p = .004$). Additionally, the indirect effect of perceived comment authenticity through trust on post-comment booking intention is also significant (effect = 0.342, $p < .001$). These results indicate that the authenticity cues mainly affect booking intention by enhancing trust.

TABLE 13 - Specific indirect effects

Indirect path	Effect	<i>t</i> -value	<i>p</i> -value	95% <i>CI</i>	Significant
Review type -> Review authenticity -> Post-review booking intention	0.049	0.692	.489	[-0.110, 0.174]	No
Review type -> Review authenticity -> Trust -> Post-review booking intention	0.166	2.902	.004	[0.076, 0.303]	Yes
Review type -> Review authenticity -> Perceived risk -> Post-review booking intention	0.021	0.571	.568	[-0.049, 0.101]	No
Review type -> Trust -> Post-review booking intention	-0.003	-0.059	.953	[-0.099, 0.085]	No
Review type -> Perceived risk -> Post-review booking intention	-0.018	-0.516	.606	[-0.097, 0.041]	No
Review authenticity -> Trust -> Post-review booking intention	0.342	3.634	< .001	[0.189, 0.556]	Yes
Review authenticity -> Perceived risk -> Post-review booking intention	0.043	0.582	.560	[-0.100, 0.196]	No

Note. Bootstrapped confidence intervals are based on 5,000 resamples.

Performance map analysis

This study conducted an IPMA analysis with post-booking intention as the target construct. In this analysis, importance denotes the total effect of each antecedent variable on post-booking intention, and performance value denotes the latent variable score, recalibrated to the 0–100 range. Table 14 and Figure 1 summarize the analysis results.

Table 14. Importance-performance map analysis for post-review booking intention

Target construct	Antecedent construct	Importance	Performance
Post-review booking intention	Perceived review authenticity	0.487	48.529
Post-review booking intention	Pre-review booking intention	0.474	46.140
Post-review booking intention	Trust	0.457	55.331
Post-review booking intention	Review type	0.128	47.059
Post-review booking intention	Perceived risk	-0.062	68.842

Note. Importance refers to the total effect on post-review booking intention. Performance values are rescaled to a 0-100 metric.

The perceived authenticity of the review has the highest total importance (0.487) on post-review booking intention, but its performance value is relatively low (48.529). The pre-review booking intention also shows high importance and low performance value, while trust has both high importance and medium performance value. These results indicate that in the current model, authenticity cues and trust-building signals are the most promising directions for improving booking intention. The total effect of perceived risk is negative and has a high-performance value, suggesting that in this specific situation, continuing to reduce perceived risk may not be as effective as strengthening authenticity and trust cues.

Overall, the results present a consistent pattern: Real review conditions enhance perceived review authenticity; perceived review authenticity further strengthens trust and reduces perceived risk; and trust and consumers' pre-review booking intention jointly explain post-review booking intention. Therefore, the results of this study support a trust-based explanation: Online review authenticity mainly affects consumers' booking decisions through the trust formation process.